Mohammadreza Zare Amarghani

Software Engineer · Backend, Databases, Automation & Applied Deep Learning

M.Sc. in Computer Engineering (Software) · Mashhad, Khorasan Razavi, Iran

- mrzroot.github.io
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Software engineer **holding a Master's degree in Computer Engineering (Software)** from Imam Reza International University, with hands-on experience in **database design and integration, backend development and business-process automation**. At the **Khorasan Razavi Chamber of Commerce, Industries, Mines and Agriculture**, reverse-engineered and documented the organisation's information systems, designed a Python ETL into a unified master database, and built the operational-budgeting system and SMS management server end to end, from schema to Windows Server deployment. For the thesis, implemented a **hybrid CNN + Vision Transformer framework in PyTorch** and evaluated it with a controlled multi-seed design, statistical significance testing and quantitative explainability checks. Maintains 15+ open-source projects in Python, TypeScript and Go, including CLI tools for developers and AI coding agents.

98.36%	**7 × 3**	**25**	**15+**
test accuracy, ResNet50–ViT	models × seeds, controlled design	systems catalogued at the Chamber	open-source projects on GitHub

EXPERIENCE

Khorasan Razavi Chamber of Commerce, Industries, Mines and Agriculture · Mashhad

Software development, databases and systems integration

Data governance & systems integration

`SQL Server · T-SQL · MySQL · Python ETL · DBML · OpenAPI 3.0 · Postman · SSO`

- Reverse-engineered and documented 9 MySQL systems and a legacy SQL Server database (100+ tables plus views, stored procedures and functions).
- Produced identity cards for 25 systems, a data catalogue and a data-sensitivity matrix as the basis for data governance.
- Designed a Python ETL into a unified master database (golden record) feeding a 360° member dashboard.
- Modelled the ERD in DBML, specified API contracts in OpenAPI 3.0 with a Postman collection, and wrote the target architecture for a unified single-sign-on (SSO) portal.

Operational budgeting system

`Laravel · Vue 3 · TypeScript · Inertia · MySQL/MariaDB · Windows Server · IIS`

- Designed and built a web budgeting system with role-based access control (system admin, manager, staff, viewer).
- Designed the schema in DBML and implemented it on MySQL/MariaDB.
- Deployed on Windows Server + IIS with an automated install script for repeatable setup.

SMS management server

`Python · Flask · Waitress · SQL Server (pyodbc) · SQLite · openpyxl · Windows service`

EDUCATION

M.Sc. in Computer Engineering (Software)

Imam Reza International University, Faculty of Engineering, Department of Computer Engineering

Defense: October 2026 (Mehr 1405)

Thesis: A hybrid framework based on convolutional neural networks and Vision Transformers for intelligent, explainable plant-disease detection

Supervisor: Dr. Kaveh Akbarzadeh Sharbaf

CORE STRENGTHS

- Database design, reverse-engineering and integration
- Backend development and contract-first APIs
- Process automation and Windows Server deployment
- Computer-vision deep learning in PyTorch
- Experimental design and significance testing
- Explainable AI (XAI)
- Tooling for developers and AI agents

- Built the organisation's SMS service on SQL Server with Excel import/export and full Jalali-calendar support.
- Finance and follow-up reporting, audit logging, admin panel and backups.
- Runs as a Windows service on Waitress; core logic covered by automated tests.

Web system for the Arbitration Centre of Iran Chamber · Mashhad office

`Vue 3 · Vite · Pinia · Django REST Framework · Token auth`

- Case filing and tracking, arbitration-fee calculator and staff workflow with a right-to-left UI; MVP delivered to the client.

RESEARCH · M.SC. THESIS

A hybrid CNN–Vision Transformer framework for explainable plant-disease detection

`PyTorch · torchvision · ResNet18/50 · ViT-B/16 · Feature fusion · Grad-CAM · Attention Rollout · SciPy`

- **Data:** PlantVillage, 54,305 images in 38 classes; removed 21 exact-duplicate groups and 35 near-duplicate pairs before splitting to prevent leakage (train 38,004 / val 8,126 / test 8,121).
- **Controlled design:** 7 models × 3 random seeds × 10 epochs with shared preprocessing and training rules, compared in two groups: trained from scratch and frozen pretrained extractors.
- **Architecture:** two-branch feature fusion; the 2048-d ResNet50 and 768-d ViT-B/16 vectors are each projected to 256 features and concatenated.
- **Result:** ResNet50–ViT reached the highest mean test accuracy (**98.36%**) and macro-F1 **97.62%** vs 96.84% for standalone ViT; the gap was significant in all three seeds (exact **McNemar** test, **Holm** correction over 33 comparisons).
- **Robustness:** on 380 balanced images, ResNet18–ViT led under heavy noise (81.40% vs 68.86%) and ResNet50–ViT under heavy blur (66.05%).
- **Ablation:** removing the ViT features dropped ResNet50–ViT from 97.46% to 87.46%; removing the CNN features, to 92.37%.
- **Explainability:** Grad-CAM and Attention Rollout with quantitative deletion/insertion evaluation and a head-randomisation sanity check; limitations reported transparently.

LANGUAGES

Persian: native

English: technical reading & writing

LINKS

`Portfolio` mrzroot.github.io

`GitHub` github.com/mrzroot

`LinkedIn` linkedin.com/in/mrzroot

`Telegram` t.me/mrzroot

SELECTED OPEN SOURCE

whystuck `Python`

Runs any command under an idle watchdog; when it hangs (pager, editor, y/N prompt, password, watch mode, lock) it says why and prints the non-interactive fix. Zero deps, JSON output for agents.

shellparity `Python`

Diffs your interactive shell against an AI agent's non-interactive shell (PATH, tool versions, proxies, UTF-8, SSH, git signing) and prints the exact rc-file fix.

whystale `Python`

Finds what is older than its source: servers started before .env changed, builds older than code, deps older than the lockfile, generated code older than its schema.

whycached `Python`

Finds which cache layer (CDN, browser, service worker, asset name) serves the stale copy of a live URL, with the fix for that layer.

persianprobe `Python`

Runs the same 14 Persian-text probes through the runtimes on your machine and shows where they disagree, with the fix for each.

whyenv `Python`

Lists every place that can set an env variable in the order your framework reads them (Vite, Next.js, CRA, dotenv, Compose), marks the winner and the ignored edit.

MihanStack `TypeScript`

Local emulator for Iranian payment and SMS APIs (Zarinpal, Zibal, Kavenegar, SMS.ir) with failure scenarios, webhooks and record/replay; 62 tests.

dibs `Python`

Guards hand-made edits against AI coding agents via git hooks and MCP; zero dependencies, 49 tests.

hamyad · agentforge `TypeScript`

Shared project memory for AI tools (MCP server on stdio and Cloudflare Workers); rule sync across Cursor, Claude Code, Windsurf and Copilot.

marz · sabr `Go`

Service-reachability scanner for sanctioned regions (WebAssembly, SARIF) and resumable git/docker downloads.

dokhaneh · flowpilot `Python`

Release mirroring to GitHub and Iranian hosts with Ed25519-signed manifests; self-hosted workflow automation with Telegram.

paknevis · netdoctor-ir `Python`

Persian text linter and auto-fixer (ZWNJ, characters, digits); connectivity doctor and mirror switcher for developers.

SKILLS MATRIX

Area	Technologies	Applied in
Programming	Python · SQL / T-SQL · TypeScript · JavaScript · Go · PHP · PowerShell	Chamber systems, thesis, open source
Databases & data	SQL Server · MySQL / MariaDB · PostgreSQL · SQLite · Redis · DBML · ETL	10 databases analysed, data catalogue, golden record
Backend & APIs	Django / DRF · Flask · FastAPI · Laravel · Node.js · REST · OpenAPI 3.0 · MCP	Budgeting system, SMS server, arbitration centre
Frontend	Vue 3 · Pinia · Inertia · React · Tailwind CSS · RTL UI	Right-to-left enterprise interfaces
Deep learning	PyTorch · torchvision · CNN (ResNet18/50) · ViT-B/16 · Transfer learning · Feature fusion	21 training runs in the thesis
Explainable AI	Grad-CAM · Attention Rollout · Deletion / Insertion · Sanity checks	Quantitative evaluation of saliency maps
Statistics & evaluation	Exact McNemar · Holm · Macro-F1 · Precision / Recall · Multi-seed · Ablation · NumPy · SciPy · Matplotlib	Significance testing of 7 models over 3 seeds
Infrastructure & tools	Git · GitHub Actions · Docker · Jenkins · Windows Server / IIS · Linux	Enterprise deployment, open-source releases
Networking	CCNA / Network+ level · Cisco Packet Tracer · MikroTik	Network labs and configuration